

BOL'SHERECHYE, Russian FederationWMO: **285930**Lat: **56.0941N**Lon: **74.6103E**Elev: **77**StdP: **100.40**Time Zone: **6.00 (E06)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.9	-32.7	-38.8	0.1	-35.7	-35.6	0.1	-32.5	8.6	-7.7	7.7	-10.2	0.8	210	0.618

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	29.4	19.3	27.6	18.6	26.0	17.8	20.9	26.8	20.0	25.5	19.0	24.0	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	13.8	23.4	17.9	13.0	22.8	17.0	12.2	21.8	60.9	26.8	57.7	25.5	54.4	24.0	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.3	6.3	-38.3	31.8	4.5	2.0	-41.6	33.2	-44.2	34.3	-46.7	35.4	-50.0	36.9		
			-38.4	22.4	4.5	0.9	-41.6	23.1	-44.2	23.6	-46.7	24.1	-50.0	24.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-18.5	-16.3	-6.7	3.3	11.4	17.1	18.2	15.7	9.9	2.7	-7.1	-15.2
	DBStd	14.34	8.19	7.55	5.78	5.98	5.38	4.42	3.69	3.94	4.46	4.94	7.39	8.21
	HDD10.0	3987	884	738	519	212	48	4	0	3	56	229	514	781
	HDD18.3	6329	1142	971	778	450	222	74	48	101	256	483	764	1039
	CDD10.0	811	0	0	0	12	92	216	256	180	52	4	0	0
	CDD18.3	111	0	0	0	0	7	37	46	19	2	0	0	0
	CDH23.3	1107	0	0	0	4	116	372	405	185	24	0	0	0
	CDH26.7	260	0	0	0	1	25	94	97	41	2	0	0	0
	WSAvg	2.8	2.5	2.8	3.1	3.2	3.1	2.6	2.3	2.4	2.6	3.0	3.2	2.9
(15) Precipitation	PrecAvg	405	18	13	16	24	38	56	75	52	33	28	30	23
	PrecMax	642	41	24	39	114	81	127	220	117	83	52	59	41
	PrecMin	280	1	0	1	1	13	10	11	10	9	7	6	8
	PrecStd	85	10	6	10	21	20	26	43	28	18	13	14	9
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.0	1.5	6.3	23.2	29.5	31.2	31.7	30.2	26.2	18.1	6.7	1.4
		MCWB	-0.7	-0.1	2.7	13.2	16.8	19.1	21.4	19.7	16.1	10.5	5.0	0.7
	2%	DB	-2.0	-0.7	4.1	18.5	26.3	29.2	29.2	27.3	22.8	15.0	4.7	0.0
		MCWB	-2.6	-1.7	1.5	10.4	15.5	18.8	20.1	18.9	14.7	9.4	3.5	-0.8
	5%	DB	-3.9	-2.5	2.6	15.5	24.0	27.4	27.4	25.3	19.8	12.3	2.8	-2.1
		MCWB	-4.5	-3.3	0.7	9.3	14.5	18.5	19.4	17.8	13.4	8.0	1.7	-2.9
	10%	DB	-6.4	-5.1	1.3	12.7	21.3	25.4	25.6	23.2	17.6	9.9	1.2	-4.3
		MCWB	-6.9	-5.9	-0.2	7.6	13.3	17.7	18.8	17.0	12.2	6.5	0.3	-4.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	0.0	3.2	13.8	17.9	22.1	22.4	21.1	17.5	11.6	5.4	1.0
		MCDB	-0.3	1.1	5.9	21.3	26.7	28.5	28.8	27.1	24.0	16.4	6.2	1.2
	2%	WB	-2.7	-1.6	1.9	11.2	16.5	20.5	21.2	20.0	15.4	10.0	3.7	-0.8
		MCDB	-2.1	-0.7	3.6	17.3	24.8	26.3	27.3	25.5	20.9	14.0	4.6	-0.1
	5%	WB	-4.5	-3.3	0.9	9.6	15.1	19.3	20.3	18.9	14.0	8.5	1.8	-2.8
		MCDB	-4.0	-2.5	2.4	15.0	22.5	25.0	25.8	23.6	19.2	11.7	2.7	-2.1
	10%	WB	-6.9	-5.9	-0.1	7.9	13.8	18.2	19.4	17.7	12.6	7.0	0.4	-4.9
		MCDB	-6.4	-5.2	1.3	12.5	20.1	23.9	24.3	22.2	16.8	9.5	1.1	-4.3
(35) Mean Daily Temperature Range	5% DB	MDBR	7.6	8.7	8.8	9.5	11.8	10.6	9.9	10.1	9.3	7.0	5.9	7.3
		MCDBR	7.5	7.1	7.9	13.9	15.6	13.9	12.8	13.4	13.1	11.3	5.6	7.0
		MCWBR	7.3	6.7	6.0	8.0	7.1	6.0	5.3	6.2	6.6	6.7	5.2	6.8
		MCDBR	7.3	7.3	6.9	13.0	14.3	11.6	11.1	11.5	11.9	10.4	5.5	6.9
	5% WB	MCWBR	7.2	6.9	5.6	7.8	7.3	5.9	5.3	6.0	6.5	6.7	5.2	6.8
(40) Clear-Sky Solar Irradiance	taub		0.280	0.313	0.372	0.410	0.401	0.410	0.428	0.423	0.368	0.353	0.316	0.283
	taud		2.036	2.014	1.923	2.076	2.232	2.283	2.230	2.217	2.332	2.220	2.076	1.978
	Ebn at Noon		586	715	757	798	837	834	807	782	780	673	528	459
	Edh at Noon		64	99	140	140	128	123	127	121	94	81	62	53
(44) All-Sky Solar Radiation	RadAvg		0.85	1.91	3.45	4.56	5.31	5.86	5.38	4.25	2.92	1.45	0.69	0.53
	RadStd		0.07	0.11	0.22	0.38	0.51	0.46	0.36	0.40	0.29	0.16	0.12	0.07

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page